Arkanil Paul

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Research Interests

My research interests lie in **computer architecture and systems**, with a focus on **cache and memory systems**, **on-chip interconnects** (including **3D NoCs**), **microarchitecture of CPU/GPU, ML/DL accelerators**, and **high-performance computing**. I work across the stack, from **RTL design in SystemVerilog** and architectural simulation in **gem5, ONNXim and GPGPU-Sim** to workload profiling and **systems software in C/C++**.

Education

Indian Institute of Technology, Roorkee — Roorkee, India
Integrated M.Tech. in Geophysical Technology; ***CGPA: 8.054/10*** — Jun 2026

Thesis: Integrating Geophysical Analysis and Machine Learning for Landslide Spatial Probability Mapping in the Garhwal Himalaya, India

Relevant Coursework

Computer Architecture and Microprocessors (CSN-221); Computer Architecture for AI (DA-408);
Discrete Mathematics (MAC-102); Fundamentals of Electronics (ECN-102); Graph Theory (MA-416)

Research Experiences

School of Electrical and Computer Sciences, IIT Bhubaneswar — Bhubaneswar, India

- *Project Associate, NPU Simulation and GPU Microarchitecture* — 08/2026 - Present
  - Working on **NPU simulation** with **ONNXim** to **integrate 3D NoC** interconnects through reusable interfaces.
  - Beginning research on **Blackwell** RTX 5000/6000 Pro **GPUs**, focusing on **cache policies** & **memory behavior**.

Department of Earth Sciences, IIT Roorkee — Roorkee, India

- *Dissertation, Landslide Spatial Probability Mapping (advised by Prof. Pallavi Chattopadhyay)* — 06/2025 - 05/2026
  - Prepared 5,621 labelled samples and 32 model features from terrain, rainfall, land-cover and geological inputs.
  - Selected XGBoost over MLP & Random Forest using **Optuna** to tune hyperparameters & **5-fold CV** to compare models, attaining **0.92 test ROC-AUC** & **87.7% landslide recall** for mapping regional landslide susceptibility.
  - Built a **4-stage Python pipeline** to align rasters, encode pixel features, refit the selected model and predict landslide susceptibility across 3.82M pixels, **exporting** georeferenced **GeoTIFF maps** with NoData handling.
  - Interpreted ERT and VLF-EM surveys at 3 landslide sites for qualitative corroboration of mapped susceptibility.

Department of Computer Science and Engineering, IIT Delhi — New Delhi, India

- *Research Intern, Cache Bandwidth Management* — 05/2025 - 08/2025
  - Implemented **LLC bandwidth management policies** for multiple cores & accelerators in **gem5 source code**.
  - **Debugged segfaults** and simulation failures **in gem5**, reducing crashes & improving simulation stability.
  - Evaluated policy performance on 6 CPU cores and 4 accelerators by **conducting SPEC & microbenchmarks**.
  - Identified bottlenecks & optimizations analysing microarchitectural metrics like IPC, miss-rate, latency, etc.

School of Electrical and Computer Sciences, IIT Bhubaneswar — Remote

- *Research Intern, Federated Learning and Microarchitectural Analysis* — 09/2024 - 04/2025
  - Developed an asynchronous **Federated Learning framework** in C++ **(5k LOC)** using **Boost** & **LibTorch**, with plug-in architecture featuring concurrent clients, on-demand model integration & lazy loading large datasets.
  - Applied **staleness-aware FedAsync** aggregation to incorporate client updates without waiting for slower clients.
  - Profiled 5+ server and client systems using **AMD uProf** and contributed to detailed microarchitectural analysis.
  - Validated framework scalability by **load testing** with **150 parallel clients** (50 per model) across 3 DL models.
- *Research Intern, CPU/GPU Workload Profiling* — 03/2024 - 07/2024
  - **Containerized** simulation environments **with Docker**, reducing setup time and eliminating configuration errors.
  - Executed **GPGPU-Sim** experiments on 7 DL models to support workload characterization & bottleneck analysis.
  - Generated 200GB+ of detailed execution traces with **Intel Pin tool** for complex workload analysis on ChampSim.

Projects

Mehta Family School of Data Science and Artificial Intelligence, IIT Roorkee Roorkee, India

- *ANN Accelerator on FPGA (Individual project, coursework DA-408)* 09/2025 - 10/2025
  - Designed **configurable neural-network RTL** (MNIST Digit Classifier) in **Verilog**; defaults to **8 MAC** units.
  - Integrated **UART** transmit and receive **state machines** for 8-bit grayscale image input and serial digit output.
  - Verified neural-network inference on a physical **Basys 3 FPGA**, attaining **95%** accuracy on MNIST test images.
  - Met **100 MHz timing** constraints in **Vivado** with 0.445 ns setup slack, using 34 block RAM tiles and 32 DSPs.

Self-directed

- *AsyncTiny (Individual Project)* 10/2023 - 01/2024
  - Developed a **C++11 (no STL) async runtime** library (**3k LOC**) featuring coroutines, futures, tasks, timers.
  - Supports cooperative scheduling, conditional sleep, task cancellation, & runs on **Arduino, ESP32**, Linux, etc.
  - Achieves ∼**200B RAM** overhead for **20 concurrent** tasks via **zero-copy** task reuse & memory efficient design.

Department of Electronics and Communication Engineering, IIT Roorkee Roorkee, India

- *Transformers (Team project, coursework ECN-102)* 03/2023 - 05/2023
  - Designed a **humanoid-to-car robot** under **Arduino** control, using 14 **servos** and 4 **DC motors** for actuation.
  - Engineered power & control **circuits** for lithium-ion batteries, **BMS**, **voltage converters** & **motor** actuation.
  - Programmed **C++ control** abstractions for 2 motor types, joints & wheels, with a **Bluetooth remote** interface.
  - Demonstrated **functional correctness** of wired electronics for the 18-motors robot design under Arduino control.

Self-directed

- *MazeRush (Team Project)* 01/2023 - 02/2023
  - Created a top-down **2D maze/action game** with 6 levels, each introducing a new map & entity for varied play.
  - Structured gameplay using **OOP** concepts & **factory classes** to reuse common behavior & spawn 6 enemy types.
  - Added **BFS pathfinding** & **raycast line-of-sight** detection for enemy navigation & pursuit across 2D tilemaps.
  - Produced a **Windows desktop build** in **Unity/C#**, bundling 2D scenes, scripts & assets into a playable game.

Department of Computer Science and Engineering, IIT Roorkee Roorkee, India

- *Configurable Set-associative Cache Simulator (Team project, coursework CSN-221)* 09/2022 - 10/2022
  - Created a set-associative cache simulator in **SystemVerilog** with **LRU**, **write-back** & **write-allocate** policies.
  - Achieved **95% hit rate** through validation on LU decomposition, matrix multiply (32x32) & quicksort workloads.
- *5-stage Pipelined Processor (Team project, coursework CSN-221)* 07/2022 - 09/2022
  - Implemented inst. fetch/decode, control, ALU, memory & branch mechanism in SystemVerilog, with testbenches.
  - Created **MIPS-like custom ISA** (consisting of **80** different inst.) and **standalone assembler** in modern C++.
  - Demonstrated **Fibonacci** (up to 47th number) and **Factorial** (up to 12!) MIPS instructions with minimal stalls.

Achievements

Global Rank **1005** out of 33,953 participants in **ICPC 2026 Online Challenge** 1 powered by Huawei Aug 2026
Global Rank **30** out of 18,557 participants in **Codechef** May Long One 2022 (Division 3) May 2022
Global Rank **26** out of 19,211 participants in **Codechef** April Long Two 2022 (Division 3) Apr 2022
JEE Advanced: All India Rank **7481** out of 141,699 candidates 2021
JEE Mains: All India Rank **9811** out of 1,019,133 candidates (**99.037** percentile) 2021

Technical Skills

Programming: C, Modern C++, C#, Python, SQL, Verilog/SystemVerilog
Libraries: C++ STL, Boost, LibTorch, Python libraries (XGBoost, Optuna)
Tools: AMD uProf, Intel Pin, gem5, ONNXim, GPGPU-Sim, ChampSim, Vivado Design Suite, Docker, Git, Linux
Hardware: FPGA (Basys 3), Arduino, ESP32

Positions of Responsibility

Programming and Algorithms Group, IIT Roorkee Roorkee, India

- *Member* 04/2022 - 07/2022
  - Hosted **open lectures** on **algorithms** & **programming** fundamentals, engaging **200+ attendees** on campus.